

# Types of Classification Error

Project: Transparent Book Recommender

## 1. Your mission

### Your learning target

I can build a labelled confusion matrix from matched reference and prediction pairs.

I can distinguish false positives from false negatives using a declared positive class.

I can check matrix totals and explain why equal accuracy can hide different errors.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

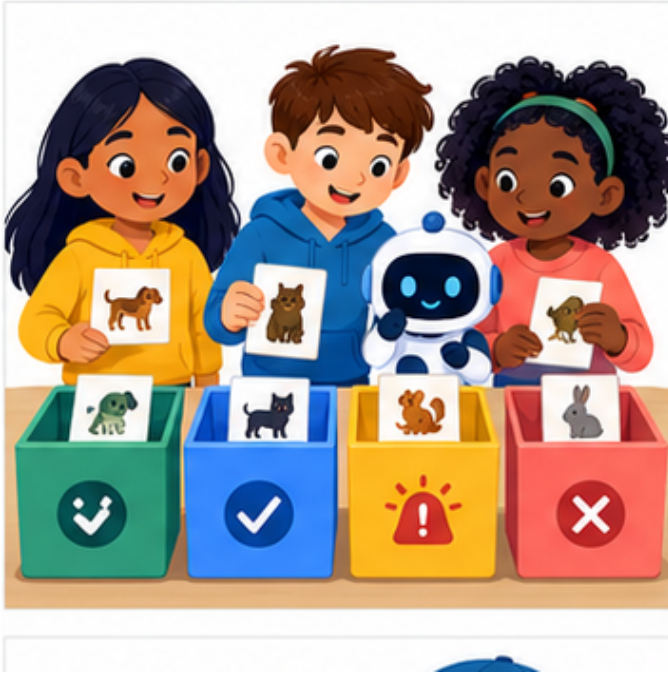
Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 13: Types of Classification Error

False alarms and misses can have different costs.



### THINKER MISSION

Sort 12 predictions into four outcome boxes.

Thinker name and date:

**Write or draw your reasoning and evidence.**

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### Readiness for the connected explanation

Why should a reproducible run save predictions linked to example IDs rather than only an accuracy score?

### This week's end goal

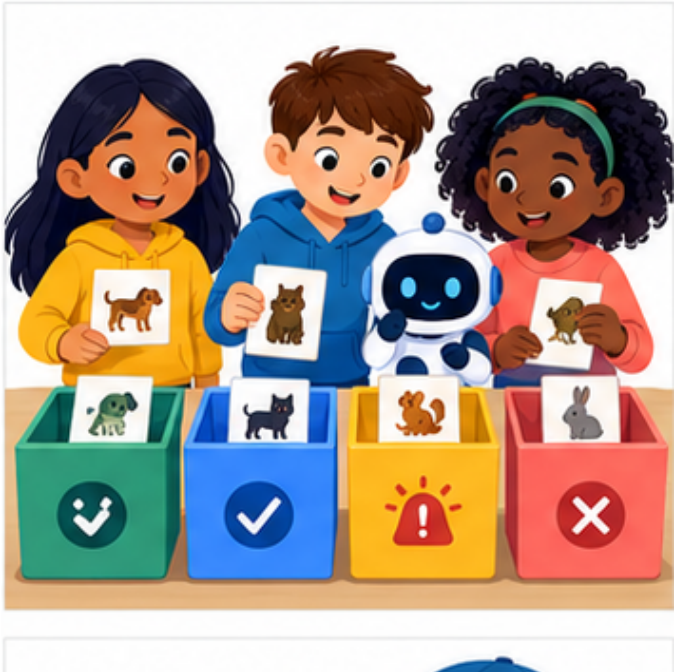
By the end, I can construct a confusion matrix with declared axes and name both error types.

## 2. Notice the evidence

### PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use types of classification error to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

**Write or draw your reasoning and evidence.**

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My prediction and the clue supporting it:

**Write or draw your reasoning and evidence.**

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## A closer explanation

A confusion matrix shows which labels a classifier mixes up. It counts examples after predictions are made; it does not fit or repair the classifier. Week 12 had reference/prediction pairs clear/clear, blocked/blocked, clear/blocked. A labelled table can preserve the type of that error as well as the total score.

This week uses eight new fictional evaluation records from a frozen classifier. They are not extra predictions from the Week 12 code. Each ID appears once with a reviewed reference and one prediction. The grouped table below lists every ID. Keep reference and prediction paired by ID before counting.

### Evaluation IDs · Reference · Prediction

**Evaluation IDs:** V1 V2

**Reference:** clear

**Prediction:** clear

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**Evaluation IDs:** V3

**Reference:** clear

**Prediction:** blocked

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**Evaluation IDs:** V4 V5

**Reference:** blocked

**Prediction:** clear

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**Evaluation IDs:** V6 V7 V8

**Reference:** blocked

**Prediction:** blocked

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Use reference labels for rows and predicted labels for columns. On both axes the order is clear, then blocked. Declare blocked as the positive class and clear as negative. Positive means the condition of interest, not good or safe. A missed blocked case and a false alarm are different errors.

## 3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

A blue rounded rectangle containing the white text "IN".

**IN**

## INPUT

Sort 12 predictions into four outcome boxes.



## PROCESS

False alarms and misses can have different costs.





OUT

## OUTPUT

Sort 12 predictions into four outcome boxes.

Which stage could fail even when the other two are correct? Explain:

**Write or draw your reasoning and evidence.**

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### A closer explanation

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pairs = [("V1", "clear", "clear"),
         ("V2", "clear", "clear"),
         ("V3", "clear", "blocked"),
         ("V4", "blocked", "clear"),
         ("V5", "blocked", "clear"),
         ("V6", "blocked", "blocked"),
         ("V7", "blocked", "blocked"),
         ("V8", "blocked", "blocked")]

labels = ["clear", "blocked"]

index = {label: i for i, label in enumerate(labels)}
```

```

counts = [[0, 0], [0, 0]]

for item_id, reference, prediction in pairs:

    counts[index[reference]][index[prediction]] += 1

print(counts)

correct = counts[0][0] + counts[1][1]

print(correct, len(pairs))

```

index maps clear to 0 and blocked to 1. Each paired record adds one to its reference row and prediction column. The script prints `[[2, 1], [2, 3]]` then `5 8`. It assumes unique IDs and only these two labels. Check IDs and missing or unresolved references before counting; the code does not validate all those conditions.

### Reference rows · Predicted clear · Predicted blocked

**Reference rows:** Reference clear

**Predicted clear:** 2

**Predicted blocked:** 1

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**Reference rows:** Reference blocked

**Predicted clear:** 2

**Predicted blocked:** 3

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The diagonal cells count correct predictions:  $2+3=5$ . Off-diagonal cells count mistakes:  $1+2=3$ . Every record must contribute exactly once. This is a count matrix, not percentages or probabilities. Keep the axis labels even when copying the numbers.

## Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

## 4. Compare the cases

### CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



**NORMAL**  
Expected everyday case



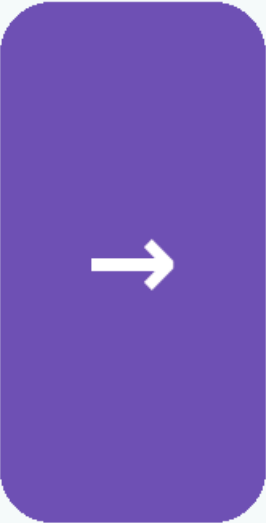
**BOUNDARY**  
Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



# MISSING

Needed information absent



# TRANSFER

Fresh situation or user

MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

**Write or draw your reasoning and evidence.**

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### Change one thing in the pictured case

Change: Change one repair/sound prediction to repair/repair after a real model revision.

Prediction: Repair row becomes [0,3]; agreement becomes 5/6.

Reason: One false negative moves to true positive; the sound row stays [2,1].

## 5. Words that unlock the idea

### VOCABULARY LAB

#### Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

## TRUE NEGATIVE

predicted no and actually  
no

3

## FALSE POSITIVE

false alarm: predicted yes,  
actually no

4

## FALSE NEGATIVE

miss: predicted no, actually  
yes

**Word · My own example · Not this / boundary**

**Word:** true positive

**My own example:** An actually positive case correctly predicted positive.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** true negative

**My own example:** An actually negative case correctly predicted negative.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** false positive

**My own example:** An actually negative case predicted positive, under a stated definition of positive.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** false negative

**My own example:** An actually positive case predicted negative, under a stated definition of positive.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Words used in the closer explanation**

## Word · Meaning

**Word:** Confusion matrix

**Meaning:** A table counting reference and prediction combinations

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**Word:** Positive class

**Meaning:** The class chosen as the condition of interest

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## 6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

### WEEK 13 • WORKED EVIDENCE

- 1 Evidence record: TP=4, TN=5, FP=2, FN=1.
- 2 Correct = TP + TN = 9; all = 12; accuracy =  $9/12 = 75\%$ .
- 3 False alarms = 2; misses = 1.
- 4 Which error matters more depends on the decision's consequence.

1. Evidence record: TP=4, TN=5, FP=2, FN=1.
2. Correct = TP + TN = 9; all = 12; accuracy =  $9/12 = 75\%$ .
3. False alarms = 2; misses = 1.
4. Which error matters more depends on the decision's consequence.

Recalculate, retrace, or restate the reasoning in your own representation:

**Write or draw your reasoning and evidence.**

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CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Blocked is the positive class

Reference and prediction · Cell name · IDs and count

Reference and prediction: clear then clear

Cell name: True negative TN

IDs and count: V1 V2 count 2

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Reference and prediction: clear then blocked

Cell name: False positive FP

IDs and count: V3 count 1

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Reference and prediction: blocked then clear

Cell name: False negative FN

IDs and count: V4 V5 count 2

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Reference and prediction: blocked then blocked

Cell name: True positive TP

IDs and count: V6 V7 V8 count 3

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Read reference first and prediction second. Here a false positive is a false blocked alarm; a false negative is a missed blocked case.

Place V3 and V4 in the matrix. Explain which is a false positive and which is a false negative when blocked is positive.

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True means prediction agrees with the reference. It does not prove that the reference annotation is infallible. Resolve disputed labels with the documented review policy rather than silently changing them to match predictions.

**Learn from the worked case**

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

**A pictorial worked case: Types of Classification Error**

A fictional book-condition checker predicts repair or sound.

**The facts we will use**

Use reference rows and prediction columns in order sound,repair. Repair is positive.

Six reference/prediction pairs: sound/sound twice; sound/repair once; repair/sound once; repair/repair twice.

AI ACADEMY / AI-2/13

**Reference rows, prediction columns**

|                  | Predicted sound | Predicted repair |
|------------------|-----------------|------------------|
| Reference sound  | 2               | 1                |
| Reference repair | 1               | 2                |

Read a cell using both its row and column labels.

## Types of Classification Error

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Declare axes

Rows: reference sound,repair. Columns: predicted sound,repair.



#### 2. Fill sound row

Two true negatives and one false positive: [2,1].



#### 3. Fill repair row

One false negative and two true positives: [1,2].



#### 4. Read result

Matrix  $[[2,1],[1,2]]$ : agreement 4/6, repair recall 2/3.

### CHANGE ONE THING

Change one repair/sound prediction to repair/repair after a real model revision.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Declare axes

**What happens and why:** Rows: reference sound,repair. Columns: predicted sound,repair.

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**Step:** 2. Fill sound row

**What happens and why:** Two true negatives and one false positive: [2,1].

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**Step: 3.** Fill repair row

**What happens and why:** One false negative and two true positives: [1,2].

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**Step: 4.** Read result

**What happens and why:** Matrix  $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ : agreement 4/6, repair recall 2/3.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

**Why this result follows**

Each case enters one cell according to its reference and predicted label. The off-diagonal cells represent different mistakes: a false alarm on a sound book versus a missed repair. Trace the cell definitions before naming rates; swapping axes changes the table’s interpretation even if its diagonal sum remains unchanged.

**What this example does not establish**

Reference labels can themselves be disputed; a true-positive name does not guarantee the annotation is infallible.

**7. Find and repair a mistake**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



**COMPARE • DIAGNOSE**

Weak idea: All mistakes should be treated as one identical error type. Strong move: locate evidence, change one cause, and retest.

**Evidence record · Expected · Actual · Likely cause / next test**

**Evidence record:** normal case

**Expected:** works

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** boundary case

**Expected:** stated rule

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** missing input

**Expected:** safe response

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

**Evidence record:** fresh transfer

**Expected:** generalizes

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_

\_\_\_\_\_

Repair one cause. Why is this repair better than changing several things?

**Write or draw your reasoning and evidence.**

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## A closer explanation

Always report the evaluated split, number of records, reference and prediction axes, label order and positive class. Reversing the axis order changes positions, not the underlying outcomes. Transposing rows

and columns swaps the two kinds of error unless the labels are corrected. Read the headers before interpreting a diagram from another source.

A zero cell means no such event in this evaluated set, not that it is impossible. A confusion matrix does not show the cause of errors, label quality or performance on future data. Keep the ID-level records for investigation. Next week examines accuracy when class counts are unequal.

**Two classifiers have accuracy 5/8. One misses two blocked records and the other misses three. Explain how both statements can be true and what must accompany their matrices for a fair comparison.**

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## 8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



### TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence\_record PREDICT expected\_result RUN one normal case and one boundary case  
COMPARE actual\_result with expected\_result IF different: locate first cause; change one factor; RETEST  
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

**Write or draw your reasoning and evidence.**

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### A closer explanation

Row totals are 3 reference-clear and 5 reference-blocked. Column totals are 4 predicted-clear and 4 predicted-blocked. All four cells sum to 8. Accuracy is the diagonal total divided by the total: 5/8, or 62.5 percent. The Week 12 three-record matrix was  $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$  in this same axis order, giving 2/3.

Another classifier on these same eight reference-labelled records could have matrix  $\begin{bmatrix} 3 & 0 \\ 3 & 2 \end{bmatrix}$ . It also gets 5/8 correct, but has zero false positives and three false negatives. Our worked classifier has one false positive and two false negatives. Equal accuracy does not settle which error pattern is preferable; consider the task and evidence before choosing.

### Practise with less help: Explain the two error types

Use workbook W2 for the connected example “Confusion matrices”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

#### Plan · Do · Check

- Plan:** What is given? Which rule or procedure applies?
- Do:** Record each intermediate step.
- Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Confusion matrices. Proposed explanation: “Positive means good”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Which error causes a sound book to be sent for unnecessary repair review?

#### My explanation using a pictured step and the given facts

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

### EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



**PREDICT → TEST → RECORD → EXPLAIN**

Place 12 actual/predicted pairs into a 2×2 grid, total each cell, check the total equals 12, and explain one error cost.

Place 12 actual/predicted pairs into a 2×2 grid, total each cell, check the total equals 12, and explain one error cost.

**Test / input · Prediction · Actual · Evidence / calculation · Change or keep?**

**Test / input:** normal

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** boundary

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** missing

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** fresh transfer

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

What did the hardest test teach you that the normal test did not?

**Write or draw your reasoning and evidence.**

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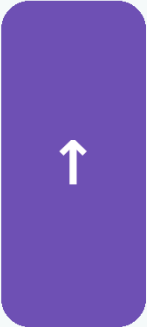
**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

MASTERY

Move upward only when you can produce evidence without copying the worked example.



### MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

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## 11. Connect your project

### CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



## TRANSPARENT BOOK RECOMMENDER

Separate recommending an unsuitable book from failing to surface a suitable one.

**THIS WEEK'S CHECKPOINT** Separate recommending an unsuitable book from failing to surface a suitable one.

**SYSTEM MAP** Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

**Write or draw your reasoning and evidence.**

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What should remain under human control?

**Write or draw your reasoning and evidence.**

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**RESPONSIBLE DESIGN** Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

**Evidence for your project**

Use a confusion table when evaluating recommender category predictions and attach concrete consequences to each error.

## 12. Show what you know

### INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



### INDEPENDENT PROOF

Sort 12 predictions into four outcome boxes.

CHALLENGE Sort 12 predictions into four outcome boxes.

1. My fresh example, trace, calculation, or design:

**Write or draw your reasoning and evidence.**

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2. My evidence that it works (include a boundary or missing-data case):

**Write or draw your reasoning and evidence.**

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3. One limitation, fairness concern, or situation needing human review:

**Write or draw your reasoning and evidence.**

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SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

## 13. Study a complete case

### Week 13 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

Positive means should recommend. Actual/predicted pairs are yes/yes, no/yes, yes/no, no/no.

#### The question

Name the four outcomes.

#### Worked explanation

True positive, false positive, false negative, true negative. Define positive before naming errors.

#### Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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**Trace the supplied case visually**

**Name outcomes using actual and predicted labels**

**Evidence or stage · Value or result**

**Evidence or stage:** True positive

**Value or result:** 1

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**Evidence or stage:** False negative

**Value or result:** 1

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**Evidence or stage:** False positive

**Value or result:** 1

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**Evidence or stage:** True negative

**Value or result:** 1

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Positive means should recommend. Actual/predicted pairs are yes/yes, no/yes, yes/no, no/no.

Name the four outcomes.

True positive, false positive, false negative, true negative. Define positive before naming errors.

## 14. Try a new case independently

**Independent case: Count a fresh evaluation independently**

Each pair below is reference then prediction: R1 clear/clear, R2 clear/blocked, R3 clear/blocked, R4 blocked/blocked, R5 blocked/clear, R6 blocked/blocked. Use clear then blocked on both axes and reference rows. Build the matrix, give row and column totals, name TP,TN,FP,FN with blocked positive, and calculate accuracy.

**My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Week 13 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Positive means sports. A sports book predicted other and a non-sports book predicted sports occur. Name each error and its consequence for the sports shelf.

My answer, with a trace, calculation or supporting evidence

**Write or draw your reasoning and evidence.**

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### Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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## Connect to your project

Separate recommending an unsuitable book from failing to surface a suitable one.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

## Transfer practice from the original programme

Positive means 'should recommend'. A book that should be recommended is rejected; a book that should not be recommended is accepted. Name both errors and explain who experiences each consequence.

**Explain the mechanism, evidence and one limit.**

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## Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.